



Evaluation of Headache

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- Headache is ache or pain in the head arising from the pain-sensitive intracranial or extracranial structures or referred from the surrounding structures.
- Headache arrange from mild to severe probably serious type.
- Headache may divided into primary, secondary or referral types.



Pain Sensitivity of Cranial Structures

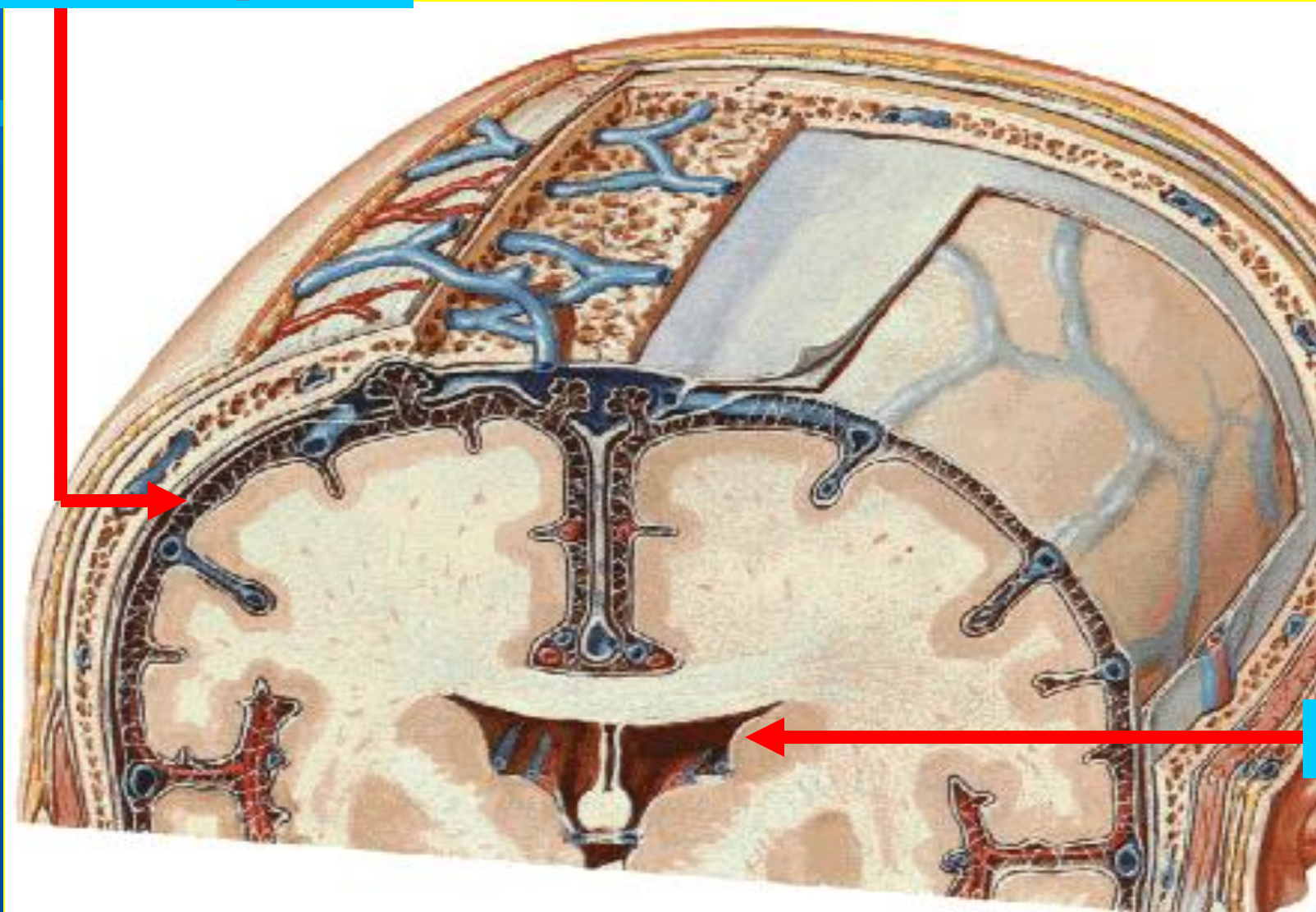
Pain-Sensitive

- Cranial **venous sinuses** with afferent veins
- **Arteries** at base of brain and their major branches
- **Arteries** of the dura
- **Dura** near base of brain and large arteries
- **Dural, Cranial and extracranial nerves**
- All **extracranial structures**

Pain-Insensitive

- Brain parenchyma
- Ependyma
- Choroid
- Pia
- Arachnoid
- Dura over convexity
- Skull

subarachnoid space



ventricles



General Mechanisms of Headache

- **Traction** on major intracranial vessels
- **Distention, dilation** of intracranial arteries
- **Inflammation** near pain sensitive structures
- **Direct pressure** on cranial or cervical nerves
- Sustained **contraction** of scalp or neck muscles
- **Stimulation** from disease of eye, ear, nose and sinuses (referred pain)



Epidemiology

- 60-75% of adults have at least one headache/year
- 5-10% will seek physician evaluation
- Less than 10% of ED patients with chief complaint of headache will have emergent secondary cause



Epidemiology

Primary Headache

- Tension
- Migraine
- Cold Stimulus Headache
- Cluster

Lifetime Prevalence

69%

15%

15%

0.1%

Secondary Headache

- Fever
- Metabolic disorder
- Disorders of nose/sinuses
- Head trauma
- Disorders of eyes
- Vascular disorders

72%

63%

22%

15%

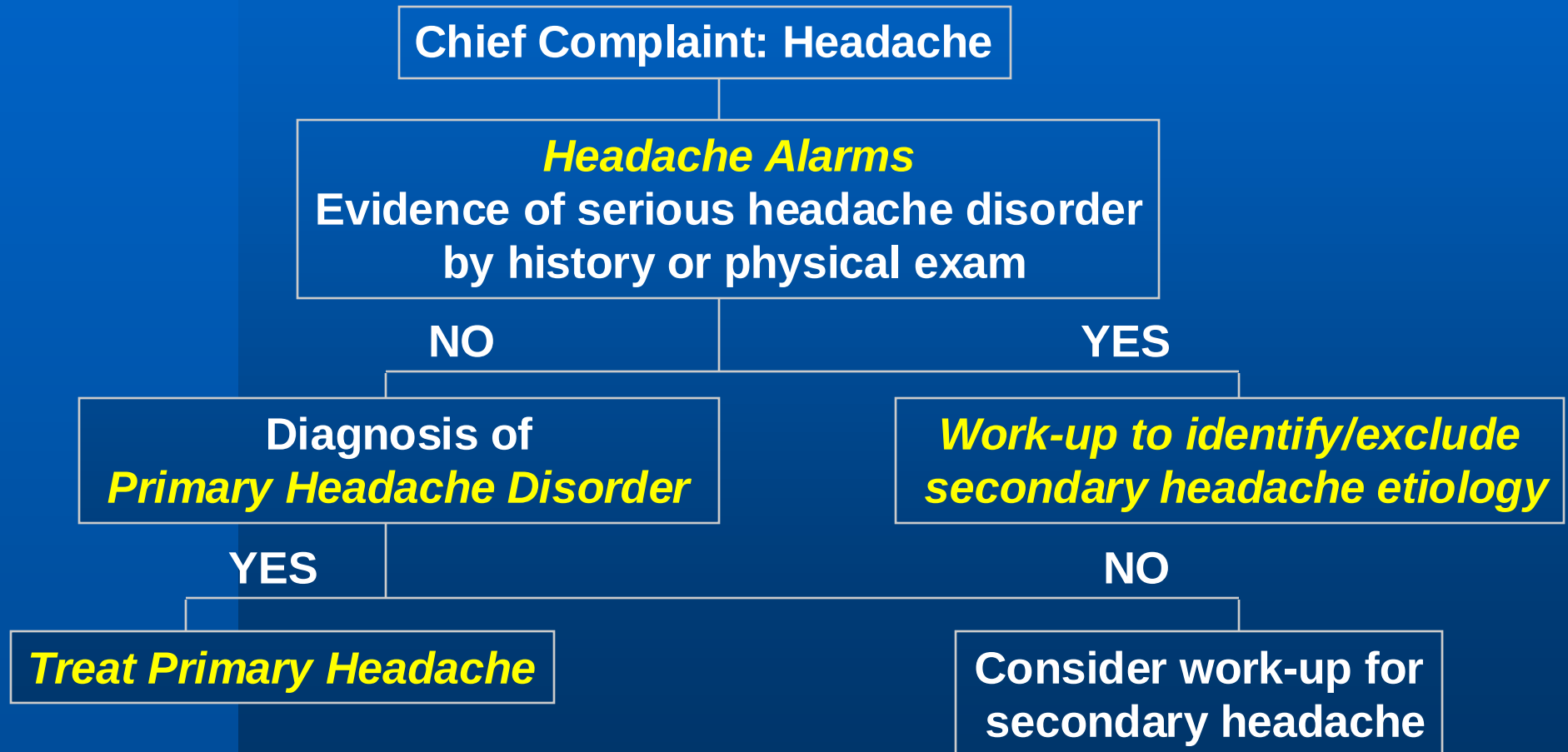
4%

3%

1%



Overall Approach





Diagnostic Alarms



- Onset after age 50
- Sudden onset
- Headache awaking patient from sleep
- Increased frequency and severity
- New onset with risk factors for HIV or cancer
- Associated with systemic illness (HT,DM,fever, meningismus, rash)
- Altered consciousness or focal neurologic deficits
- Papilledema
- Significant trauma



Primary Headache

Tension

Migraine



Cluster



Tension Headache

International Headache Society Diagnostic Criteria

- **Duration**
 - 30 min to 7 days
- **Pain characteristics** (at least 2)
 - Pressing/tightening quality
 - Mild to moderate severity
 - Bilateral location
 - No aggravation by routine physical activity
- **Associate symptoms** (Must have both)
 - No vomiting
 - No more than one of: nausea, photophobia, phonophobia
- H&P and diagnostic tests do not suggest underlying organic disease



Migraine Without Aura

International Headache Society Diagnostic Criteria

- **At least 6 or more periodic attacks**
- **Duration**
 - 4-72 h if untreated or unsuccessfully treated
- **Pain characteristics** (at least 2)
 - Unilateral location
 - Pulsating quality
 - Moderate to severe intensity
 - Aggravation by walking stairs or similar physical activity
- **Associated symptoms** (at least 1)
 - Nausea, vomiting, or both
 - Photophobia or phonophobia
- H&P and diagnostic tests do not suggest underlying organic disease



Migraine With Aura

International Headache Society Diagnostic Criteria

- **At least 3 periodic attacks**
- **Aura characteristics** (At least 3)
 - One or more fully reversible aura symptoms indicating focal cerebral cortical or brain-stem dysfunction
 - At least 1 aura symptom develops gradually over >4 minutes or 2 or more symptoms occur in succession
 - No single aura symptom lasts > 60 minutes
 - Headache begins within 60 minutes of aura onset
- History, physical, and diagnostic tests do not suggest underlying organic disease



Cluster Headache

International Headache Society Diagnostic Criteria

- **Duration**
 - 15 to 180 minutes untreated
- **Pain characteristics**
 - Severe unilateral orbital, supraorbital, or temporal pain
- **Associated symptoms** (at least 1, ipsilateral to pain)
 - Conjunctival injection, Lacrimation
 - Nasal congestion, Rhinorrhea
 - Forehead and facial swelling
 - Miosis, Ptosis
 - Eyelid Edema
- **Frequency:**
 - between 1 every other day to 8/day



Secondary Headache

- Intracranial hemorrhage
 - Subarachnoid Hemorrhage
 - Intracerebral hemorrhage
 - Subdural/epidural hematoma
- Meningitis/encephalitis
- Hypertensive encephalopathy
- Ischemic stroke
- Venous sinus thrombosis
- Hypoxia, hypercarbia, carbon monoxide



Secondary Headache

- Temporal arteritis
- Mass lesions
 - Tumor, abscess, arteriovenous malformation
- Altitude sickness
- Metabolic
 - Hypoglycemia, fever, hypothyroid, anemia
- Glaucoma
- Pseudotumor cerebri (benign intracranial hypertension)

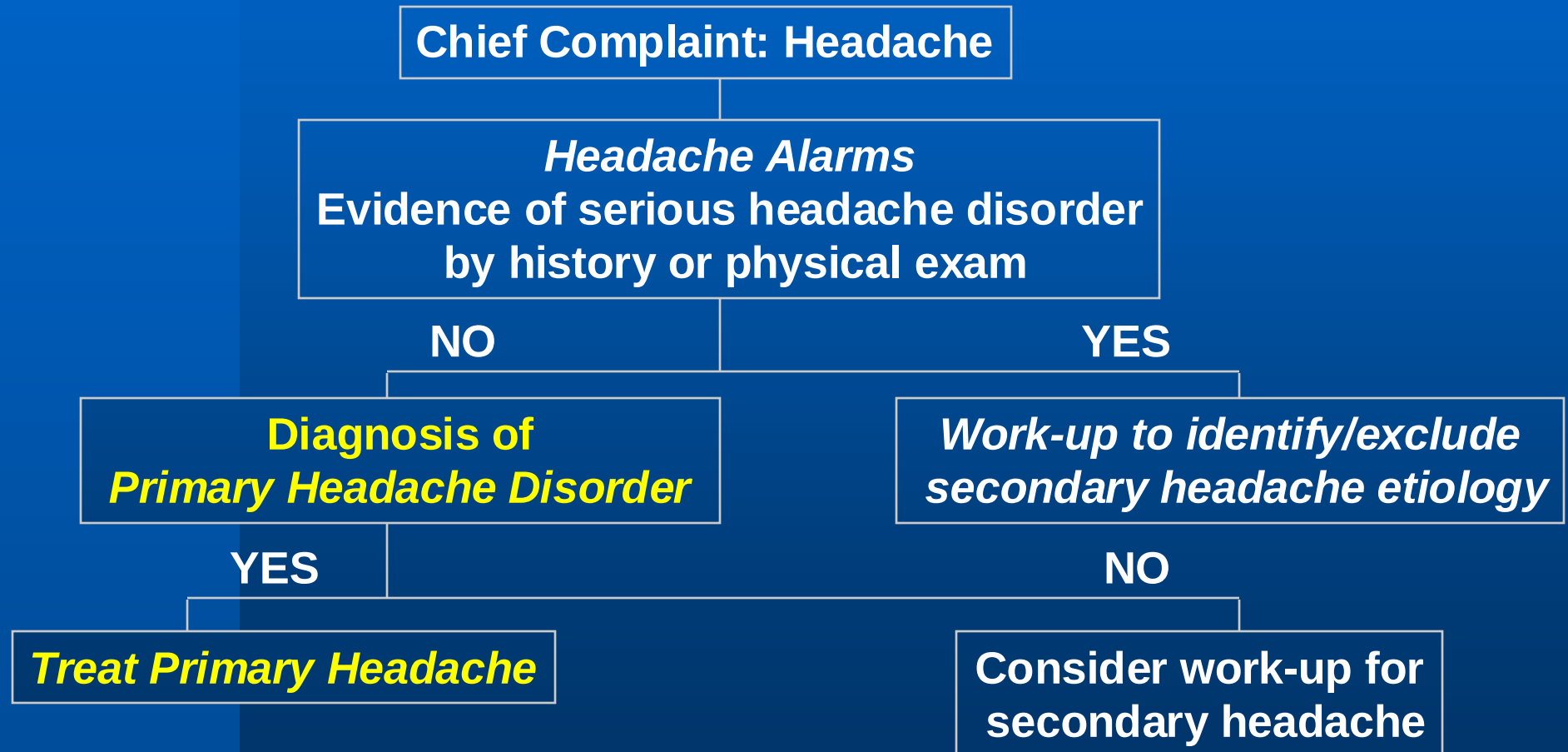


Secondary Headache

- Trigeminal Neuralgia
- Post-concussion syndrome
- Sinusitis without complication
- Post-lumbar puncture
- Diet
- Medications
- Fatigue, postexertion, postcoital



Overall Approach





ED Treatment of Primary Headache

- Tension
 - Oral Analgesics (NSAIDS, Acetaminophen)
- Migraine
 - Serotonin agonists
 - ie, Sumatriptan 50 mg PO or 6.0 mg SQ
 - Narcotics IV or IM
- Cluster
 - 100% oxygen
 - Intranasal lidocaine ?
 - NSAIDS
 - Migraine specific therapies

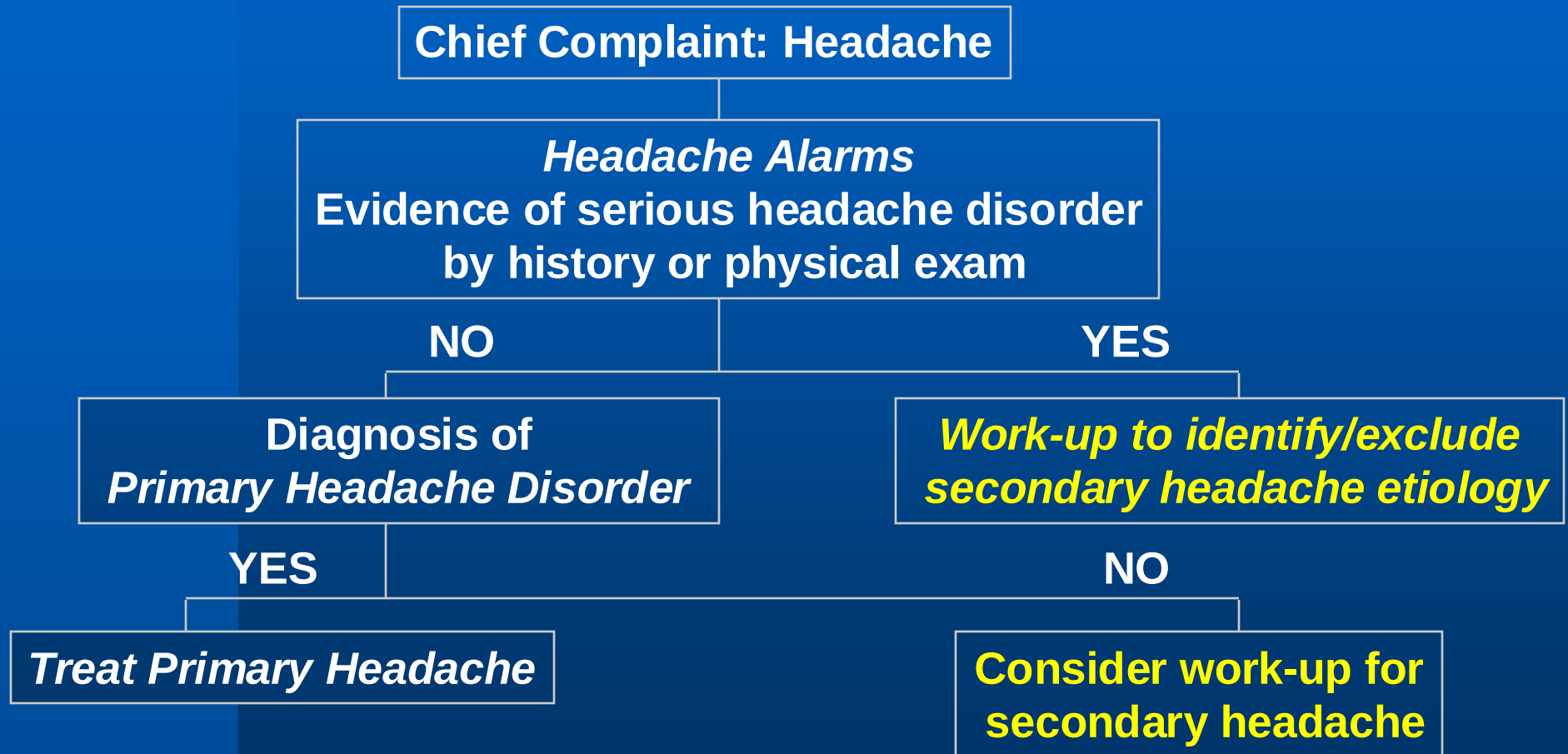


Prophylaxis Treatment of Primary Headache

- Tension
 - Reassurance
 - Antidepressant &/or Anxiolytic drugs (Tricyclic antidepressant or/& SSRI)
- Migraine
 - Betablockers: Propranolol
 - Ca channel blocker: Verapamil
 - Antidepressant: (Tricyclic antidepressant or/& SSRI)
 - Anticonvulsant: Na valproate, topiramate,
 - Methysergid.
 - Pizotifin.
- Cluster
 - Steroid
 - Lithium carbonate
 - Verapamil



Overall Approach



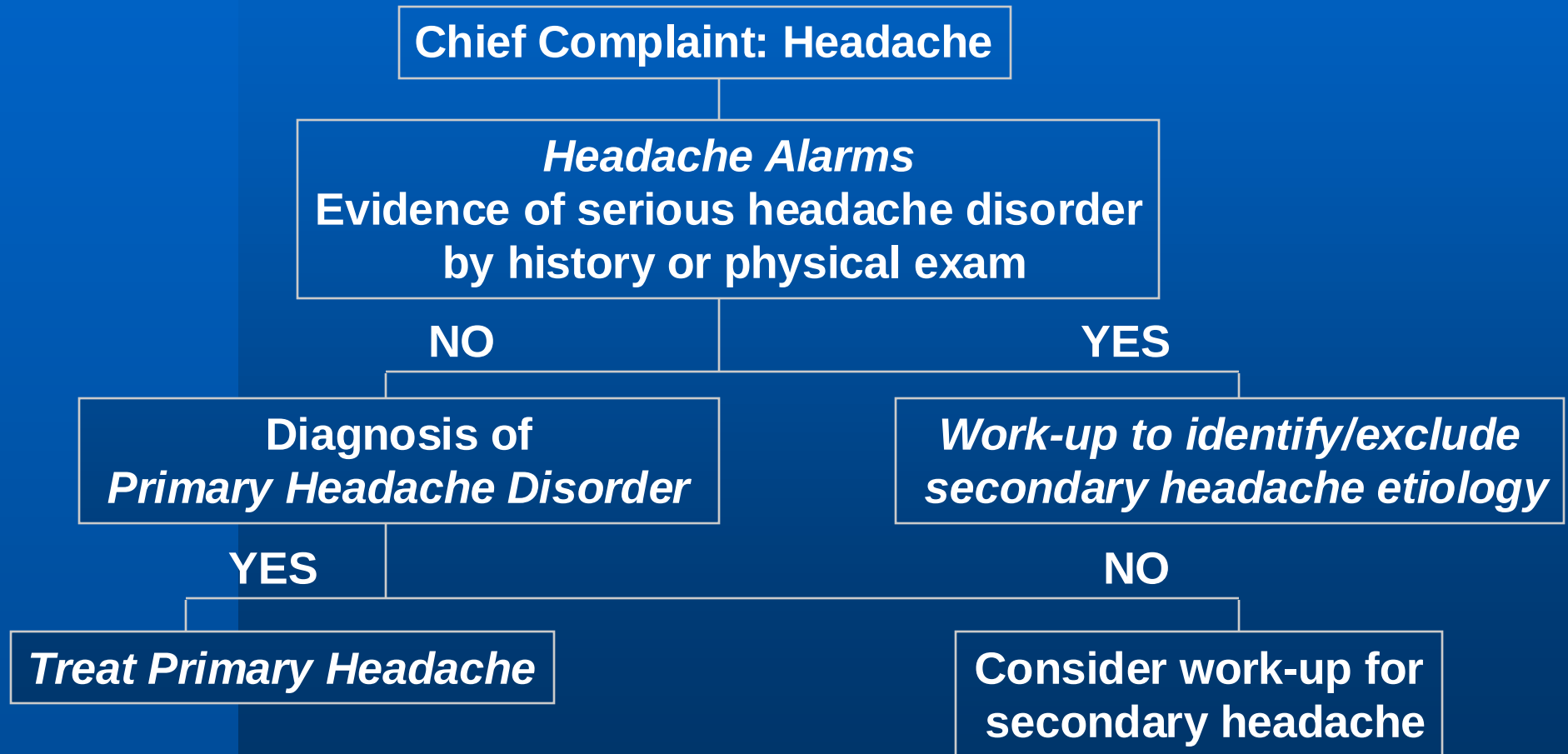


Diagnostic Studies

- Computerized tomography
 - Hemorrhage, tumor, abscess, AVM
- Lumbar puncture
 - Hemorrhage, infection, increased CSF pressure
- Limited indications for MRI, MRA, or Angiography
- Laboratory studies based on suspected etiologies
 - ESR: Temporal arteritis
 - Carboxyhemoglobin: Carbon monoxide



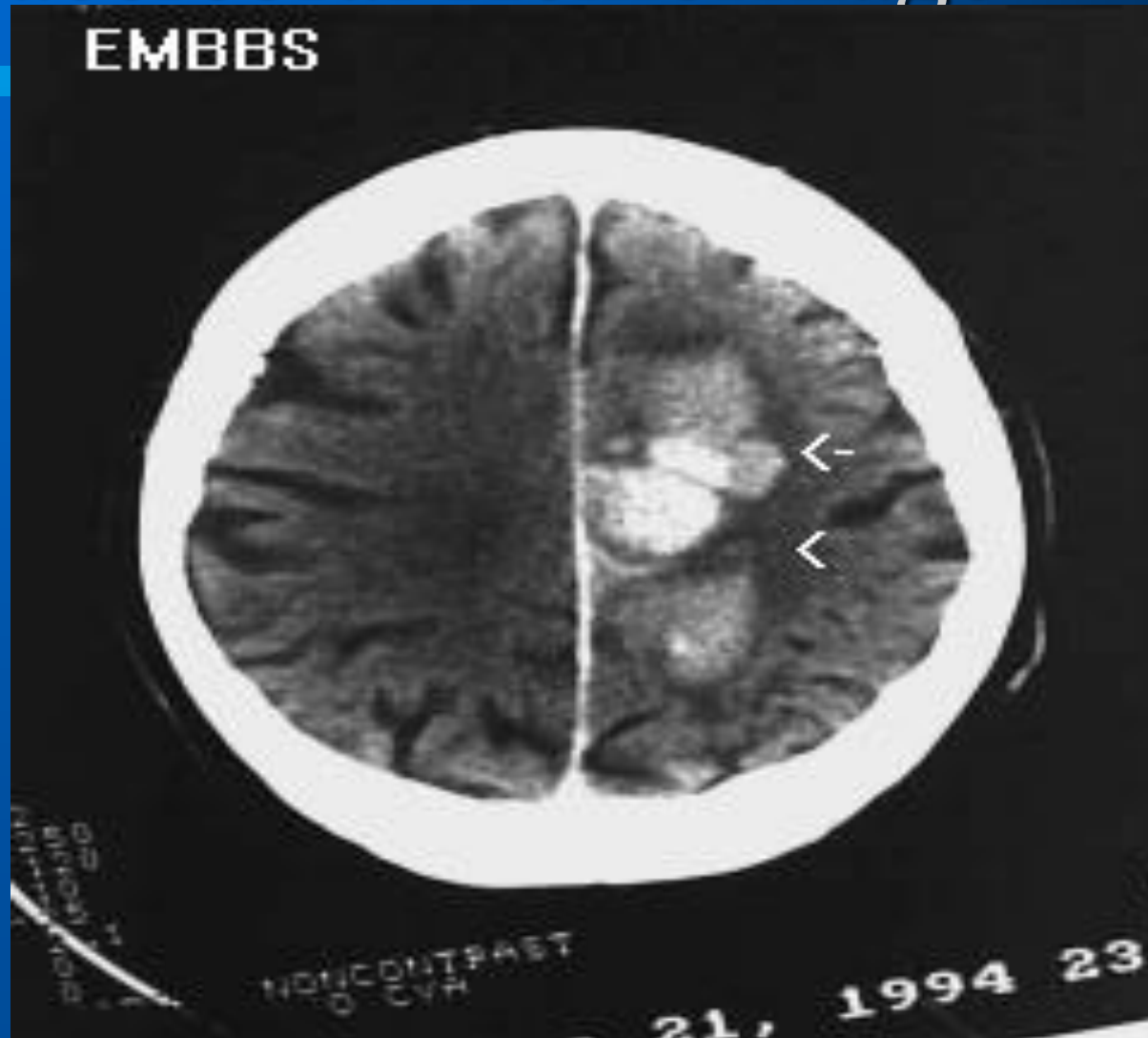
Summery



EMBBS



Intracerebral Hemorrhage



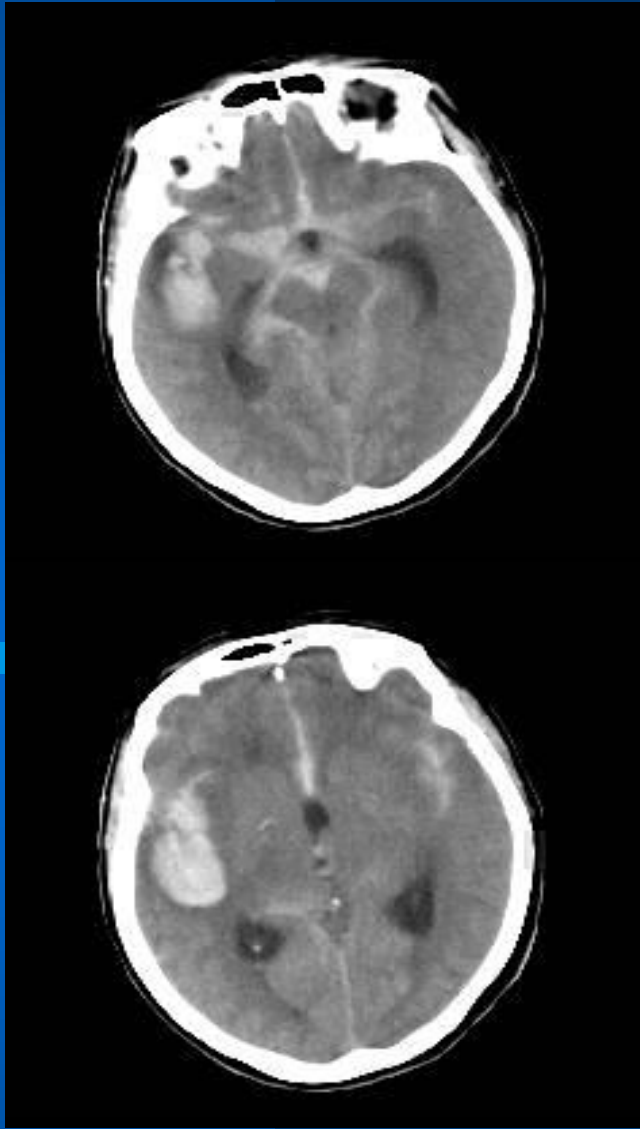
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Subdural hematoma



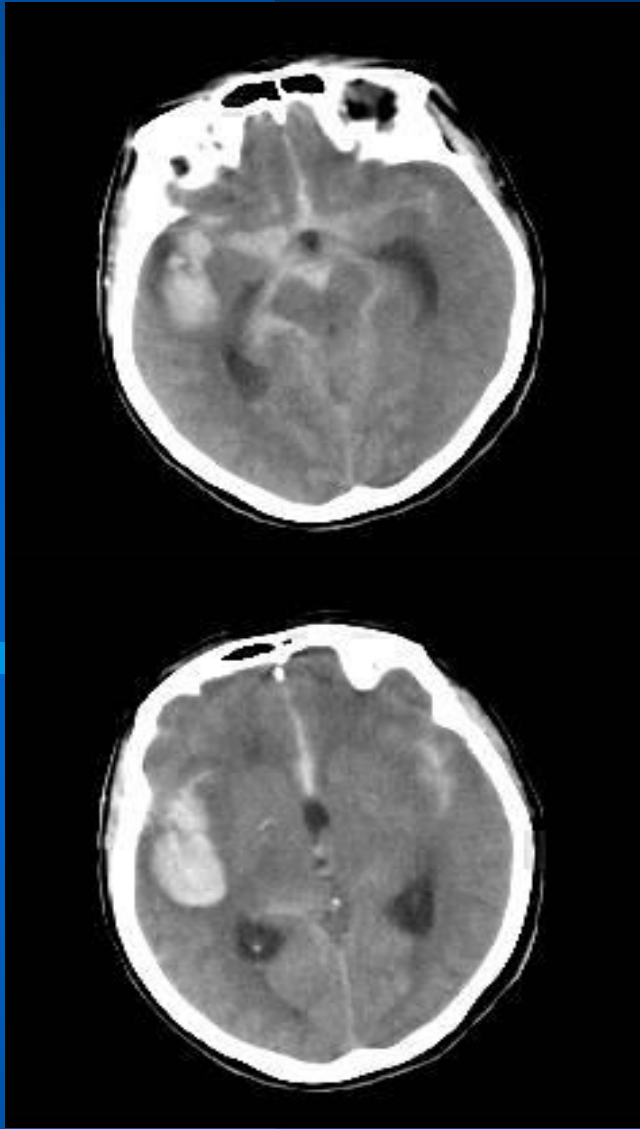
where is the aneurysm?

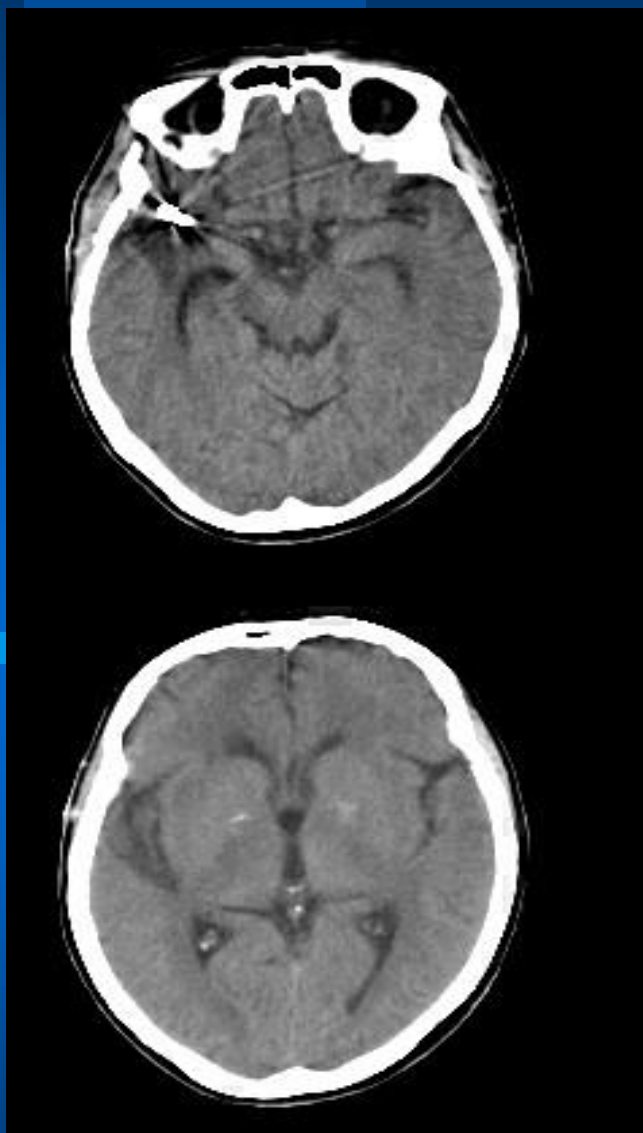


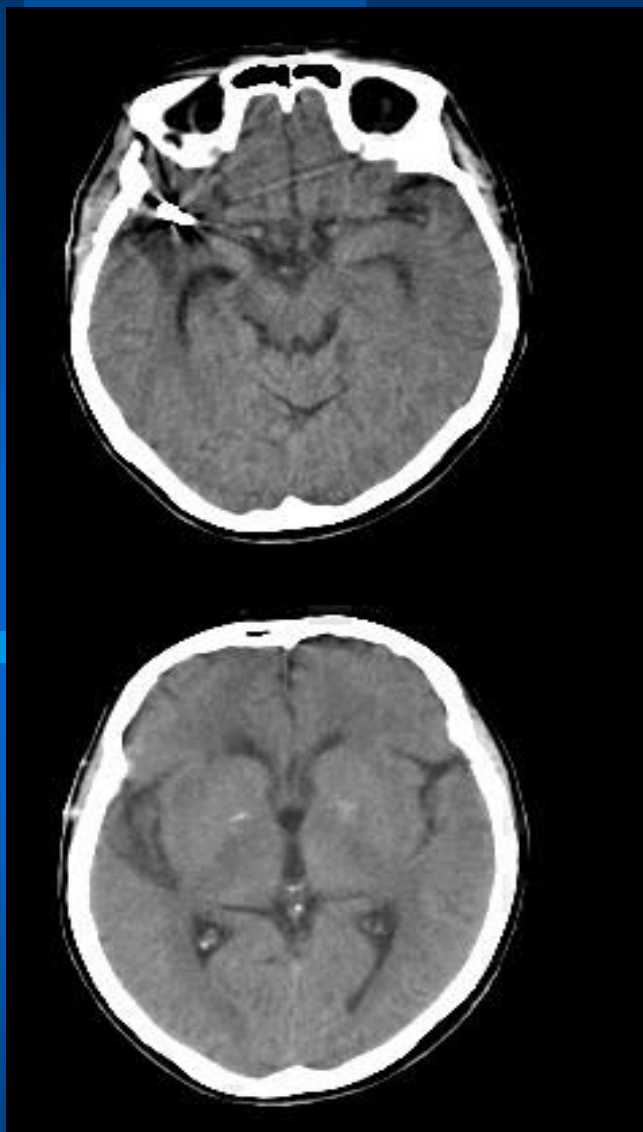
SAH



where is the aneurysm?







surgical clip



before



after

THE INCORRECT STATEMENT

- a) MIGRAIN IS EPISODIC HEADACHE
- b) CLUSTER HEADACHE IS UNCOMMON
- c) PAPILOEDEMA IS RED FLAG IN MALIGNANT HEADACHE
- d) MRI IS MANDATORY IN ACUTE ONSET HEADACHE